BRAINWARE UNIVERSITY
School of Engineering
Department of Computer Science & Engineering – Cyber Security & Data Science
398, Ramkrishnapur Road, Barasat, North 24 Parganas, Kolkata - 700 125

Review-III (A.Y.-2026-27)

Project-I, PROJ-CSD782 | Program: B.Tech. CSE-DS [Batch: 2023-27]

S.N.	Roll No.	Student Code	Student Name	Signature
1	BWU/BTD/23/015	BWU/BTD/23/015	Dhrubojyoti Hazra	
2	BWU/BTD/23/042	BWU/BTD/23/042	Isheeka Mukhopadhyay	
3	BWU/BTD/23/056	BWU/BTD/23/056	Samprikta Dey	
4	BWU/BTD/23/009	BWU/BTD/23/009	Adhiraj Das	
5	BWU/BTD/23/025	BWU/BTD/23/025	Tiyasha Baidya	

Project Title: **Fraud Detection in Digital Payment Systems Using Machine Learning**

Build a fraud detection framework capable of analyzing large volumes of digital transaction data stored in a data warehouse. Machine learning and anomaly detection techniques identify suspicious transactions in real time. The system generates risk scores and alerts to help financial institutions prevent fraudulent activities.

Name & Signature of Project Supervisor
Prof. Rudranath Mitra

Dr. Ranadip Kundu
Project Coordinator CSE – CS & DS

Project Progress (Progress up to Review 3):

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1. End-to-End System Architecture and Cloud Deployment

The team successfully engineered and deployed a production-grade multi-tier digital payment fraud detection platform across modern cloud infrastructure:

- Frontend Client (Vercel): Deployed an interactive Next.js 14 console featuring Branch Analyst Workbench, Senior Escalation Queue with 15-minute countdown SLA watchdogs, and Team Lead Executive View.
- Backend Scoring Service (Render): Deployed an asynchronous FastAPI microservice with atomic Redis idempotency achieving 11.57ms mean scoring latency, hosting calibrated XGBoost, PyTorch Autoencoder, and TreeSHAP explainability.
- Operational Database (Supabase Cloud): Configured hosted PostgreSQL with real-time WebSocket subscriptions and strict 3-tier Row-Level Security (RLS) enforced via SECURITY DEFINER functions across 5 Indian banking hubs.
- Columnar Warehouse (DuckDB): Integrated an in-process OLAP engine directly aggregating 6,362,620 transactions (Rs 1.14 Trillion turnover, 8,213 frauds) in sub-120ms without external server overhead.

2. Technical Innovations and Synthetic Leakage Remediation

Building on mid-term guidance from Prof. Rudranath Mitra, two core scientific breakthroughs were implemented:

- Leakage-Free Feature Engineering: Eliminated PaySim simulator balance leakage (raw balance deltas causing artificial 1.00 PR-AUC) by engineering pure behavioral signals (balance_drain_ratio, 1h/24h velocity, and transfer-to-cashout flags), achieving a robust, defensible PR-AUC of 0.884 and 94.21% fraud recall.
- Disjunctive Override Decision Gating: Formulated a non-linear aggregation engine where high-confidence XGBoost fraud (P_XGB >= 0.90) or severe Autoencoder novelty spikes (S_AE >= 0.95) immediately bypass linear averages directly into ESCALATE.
- Whitebox Explainability & Graph Intelligence: Integrated sub-12ms TreeSHAP attributions and 1-hop "Igor" mule ring ego-graphs for regulatory compliance (RBI guidelines / GDPR).

3. Empirical Validation and Working Module Demonstrations

All core software modules have been fully verified across 12/12 stress test scenarios in the test suite:

- Attack Simulation Sandbox: Live injection of Legitimate, Account Takeover, Zero-Day Outlier, and Blacklisted Mule vectors demonstrating real-time WebSocket triage and triage actions (APPROVE, FLAG, ESCALATE, BLOCK).
- Dual-Model Performance: Supervised XGBoost captures high-velocity drainage patterns, while the unsupervised Autoencoder isolates unlabelled zero-day anomalies by reconstruction error (MSE > tau).
- Delayed Feedback Protection: Implemented a 60-day observation blackout window in model fine-tuning to prevent premature retraining on unconfirmed chargeback labels.

4. Team Allocation, Project Deliverables, and Viva Preparedness

- Team Allocation: Dhrubojyoti Hazra (Lead, Architecture & Vercel UI), Isheeka Mukhopadhyay (ML Pipeline & Leakage Auditing), Samprikta Dey (FastAPI Backend & Rules Aggregator), Adhiraj Das (Explainable AI & Live Simulator), Tiyasha Baidya (DuckDB Warehouse & Supabase Database Security).
- Comprehensive Deliverables: 42-page technical report, reproducible Google Colab benchmark notebook, cloud-deployed web application, and full GitHub codebase (github.com/dhrubojyotihazra/ASTID).
- Viva Voce Defense Protocol: Structured defense protocol rehearsed covering PR-AUC trade-offs, concept drift detection (PSI < 0.10), delayed feedback protection, and cost-sensitive Bayes decision boundaries.